

1. $X, Y \sim N-X$

a) $N \sim \text{Pois}(\lambda)$, $X|N=n \sim \text{Bin}(n, p)$

$$P(X=x, Y=y) = \sum_n P(X=x, Y=y, N=n) = P(X=x, Y=y, N=x+y)$$

$$P(X=x, Y=y) = P(X=x, Y=y | N=n) \cdot P(N=n) = P(X=x, Y=y | N=x+y) \cdot P(N=x+y)$$

$$= \binom{x+y}{x} p^x (1-p)^y \cdot \frac{e^{-\lambda} \lambda^{x+y}}{(x+y)!} = \frac{(x+y)!}{x! y!} p^x (1-p)^y \frac{e^{-\lambda} \lambda^{x+y}}{(x+y)!}$$

$$= \frac{e^{-\lambda} \lambda^x}{x!} p^x (1-p)^y = \left(\frac{e^{-\lambda p} \lambda^x p^x}{x!} \right) \left(\frac{e^{-\lambda(1-p)} \lambda^y (1-p)^y}{y!} \right)$$

b) $P(X=x, Y=y) = P(X=x)P(Y=y)$

$\therefore X$ and Y are independent

$$P(X=x) = \sum_y P(X=x, Y=y) = \frac{e^{-\lambda p} \lambda^x p^x}{x!}$$

$$P(Y=y) = \sum_x P(X=x, Y=y) = \frac{e^{-\lambda(1-p)} \lambda^y (1-p)^y}{y!}$$

2. Gubner Chapter 2: 25

$$P_{XY}(i, j) = \begin{cases} \frac{3^{j-1} e^{-3}}{j!}, & i=1, j \geq 0 \\ \frac{4 \cdot 6^{j-1} e^{-6}}{j!}, & i=2, j \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$P_X(i=1) = \sum_{j \geq 0} P_{XY}(i=1, j) = \sum_{j \geq 0} \frac{3^{j-1} e^{-3}}{j!} = \frac{1}{3e^3} \sum_{j \geq 0} \frac{3^j}{j!} = \frac{e^3}{3e^3} = \frac{1}{3}$$

$$P_X(i=2) = \sum_{j \geq 0} P_{XY}(i=2, j) = \frac{2}{3} \cdot \frac{1}{e^6} \sum_{j \geq 0} \frac{6^j}{j!} = \frac{2}{3} \cdot \frac{e^6}{e^6} = \frac{2}{3}$$

$$P_Y(j) = \sum_{i=1}^2 P_{XY}(i, j) = \frac{3^{j-1} e^{-3}}{j!} + 4 \cdot \frac{6^{j-1} e^{-6}}{j!} = \frac{1}{3e^3} \frac{3^j}{j!} + \frac{2}{3e^6} \frac{6^j}{j!}$$

$$= \frac{1}{3} \left(\frac{e^{-3} 3^j}{j!} \right) + \frac{2}{3} \left(\frac{e^{-6} 6^j}{j!} \right)$$

$$P_X(i=1)P_Y(j) = \frac{1}{3} \left[\frac{1}{3} \left(\frac{e^{-3} 3^j}{j!} \right) + \frac{2}{3} \left(\frac{e^{-6} 6^j}{j!} \right) \right]$$

$$P_{XY}(i=1, j) = \frac{1}{3} \left(\frac{3^{j-1} e^{-3}}{j!} \right)$$

$$\therefore P_X(i=1)P_Y(j) \neq P_{XY}(i=1, j)$$

X, Y are not independent.

3. Gubner Chapter 3, 7

$X_1, X_2, \dots, X_n \sim \text{i.i.d. Geometric}(p)$

$Y = X_1 + X_2 + \dots + X_n$. Find $E(Y)$, $\text{Var}(Y)$, $E(Y^2)$.

$$P(X_i = k) = p \cdot (1-p)^{k-1}$$

$$E(X_i) = \sum_{k=1}^{\infty} k \cdot P(X_i = k) = p \sum_{k=1}^{\infty} k \cdot (1-p)^{k-1} = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

$$\text{Var}(X_i) = E(X_i^2) - E(X_i)^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$$

$$\frac{p}{1-p} \sum_{k=1}^{\infty} k (1-p)^k = \frac{p}{1-p} \left(\frac{1-p}{1-(1-p)} + \frac{(1-p)^2}{1-(1-p)} + \dots \right)$$

$$= \frac{1}{1-(1-p)} = \frac{1}{p}$$

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$$E(X_i^2) = p \sum_{k=1}^{\infty} k^2 (1-p)^{k-1}$$

$$= p \cdot \frac{1+(1-p)}{p^3} = \frac{2-p}{p^2}$$

$$E(Y) = n \cdot E(X_i) = n \cdot \frac{1}{p} = \frac{n}{p}, \quad \text{Var}(Y) = n \cdot \text{Var}(X_i) = n \cdot \frac{1-p}{p^2} \quad \because X_i \text{ are independent}$$

$$E(Y^2) = \text{Var}(Y) + E(Y)^2 = \frac{n-np}{p^2} + \frac{n^2}{p^2} = \frac{n^2 + n - np}{p^2}$$

4. X : random variable that takes integer values and symmetric

$$P(X=k) = P(X=-k) \quad \forall k \in \mathbb{Z}$$

Find Expected values of $Y_1 = \cos(\pi X)$ and $Y_2 = \sin(\pi X)$

$$Y_1 = \cos(\pi X) = (-1)^k$$

$$Y_2 = \sin(\pi X) = 0$$

$$Y_2 = \sin(\pi X) = 0$$

$$E(Y_1) = \sum_{k \in \mathbb{Z}} P(X=k) (-1)^k$$

$$E(Y_2) = \sum_{k \in \mathbb{Z}} P(X=k) \cdot 0 = 0$$

5. Gubner Chapter 3, Problem 24.

integer valued random variables X, Y

Conditioned on $Y=j$, $X \sim \text{Pois}(\lambda_j)$, Evaluate $P(X > 2 | Y=j)$

$$P(X=x | Y=j) = \frac{e^{-\lambda_j} \lambda_j^x}{x!}$$

$$P(X > 2 | Y=j) = 1 - P(X \leq 2 | Y=j)$$

$$P(X > 2 | Y = j) = 1 - P(X=0 | Y=j) - P(X=1 | Y=j) - P(X=2 | Y=j)$$

$$= 1 - e^{-\lambda_j} - e^{-\lambda_j} \lambda_j - \frac{e^{-\lambda_j} \lambda_j^2}{2!}$$

$$= 1 - e^{-\lambda_j} \left(1 + \lambda_j + \frac{1}{2} \lambda_j^2 \right)$$

6. $X_1, X_2, \dots, X_n$ are independent, $\text{Geo}(p)$, random variables.

compute $P(\min(X_1, X_2, \dots, X_n) > k)$ and $P(\max(X_1, X_2, \dots, X_n) \leq k)$

$$P(X_i = k) = (1-p)^k \cdot p$$

$$\begin{aligned} P(\min(X_1, X_2, \dots, X_n) > k) &= P(X_1 > k, X_2 > k, \dots, X_n > k) = P(X_i > k)^n = [1 - P(X_i \leq k)]^n \\ &= \left(1 - \sum_{j=0}^k P(X_i = j) \right)^n = \left(1 - \frac{1 - (1-p)^{k+1}}{1 - (1-p)} \cdot p \right)^n \\ &= (1-p)^{(k+1)n} \end{aligned}$$

$$\begin{aligned} P(\max(X_1, X_2, \dots, X_n) \leq k) &= P(X_1 \leq k, X_2 \leq k, \dots, X_n \leq k) = P(X_i \leq k)^n \\ &= \left(\sum_{j=0}^k P(X_i = j) \right)^n = (1 - (1-p)^{k+1})^n \end{aligned}$$

7. G_1, G_2, G_3 be independent geometric random variables with same pdf :

$$P_{G_i}(k) = P_{G_j}(k) = P_{G_l}(k) = p(1-p)^{k-1}, \text{ what is } P(G_1 = k | G_1 + G_2 + G_3 = n)$$

$$k = 1, 2, 3, \dots$$

about $P(G_1 = g_1, G_2 = g_2, G_3 = g_3)$ for $g_1 + g_2 + g_3 = n$ we can think probability

at (g_1) th has 1st success, $(g_1 + g_2)$ th has 2nd success and $(g_1 + g_2 + g_3 = n)$ th 3rd success

$$\begin{aligned} \therefore P(G_1 = k | G_1 + G_2 + G_3 = n) &= \frac{P(G_1 + G_2 + G_3 = n, G_1 = k)}{P(G_1 + G_2 + G_3 = n)} = \frac{\binom{n-1-(k-1)+1}{1}}{\binom{n-1}{2}} = \frac{n-k-1}{(n-1)(n-2)} \cdot 2 \\ &= \frac{2(n-k-1)}{(n-1)(n-2)} \end{aligned}$$

8. Gubner Chapter 3, Problem 26

X, Y are independent with $X \sim \text{geometric}(p)$ and $Y \sim \text{geometric}(q)$

$T = X - Y$ and find $P(T=n)$ for all n

$$P(X=x) = p^x(1-p) \quad x=0,1,2,\dots \quad P(Y=y) = q^y(1-q) \quad y=0,1,2,\dots$$

$$P(T=n) = \begin{cases} \sum_{k=0}^{\infty} P(X=k+n, Y=k) & n \geq 0 \\ \sum_{k=-n}^{\infty} P(X=k+n, Y=k) & n < 0 \end{cases}$$

$$\begin{aligned} \text{if } n \geq 0: \sum_{k=0}^{\infty} P(X=k+n, Y=k) &= \sum_{k=0}^{\infty} P(X=k+n)P(Y=k) = \sum_{k=0}^{\infty} p^{k+n}(1-p) \cdot q^k(1-q) \\ &= p^n(1-p)(1-q) \sum_{k=0}^{\infty} (pq)^k = \frac{p^n(1-p)(1-q)}{1-pq} \end{aligned}$$

$$\begin{aligned} n < 0: \sum_{k=-n}^{\infty} P(X=k+n, Y=k) &= \sum_{k=-n}^{\infty} P(X=k+n)P(Y=k) = \sum_{k=-n}^{\infty} p^{k+n}(1-p)q^k(1-q) \\ &= p^n(1-p)(1-q) \sum_{k=-n}^{\infty} (pq)^k = p^n(1-p)(1-q) \frac{(pq)^{-n}}{1-pq} \\ &= \frac{q^n(1-p)(1-q)}{1-pq} \end{aligned}$$

9. Gubner Chapter 3, Problem 30

$X \sim \text{Pois}(x)$, $Y|X=n \sim \text{Ber}(\frac{1}{n+1})$

Find $P(X=n|Y=1)$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad P(Y=0|X=n) = \frac{P(Y=0, X=n)}{P(X=n)} = 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

$$P(Y=1|X=n) = \frac{P(Y=1, X=n)}{P(X=n)} = \frac{1}{n+1}$$

$$P(X=n|Y=1)P(Y=1) = P(Y=1|X=n)P(X=n)$$

$$\therefore P(X=n|Y=1) = \frac{P(Y=1|X=n)P(X=n)}{P(Y=1)}$$

$$P(Y=1) = \sum_{n=0}^{\infty} P(X=n, Y=1) = \sum_{n=0}^{\infty} P(Y=1|X=n)P(X=n) = \sum_{n=0}^{\infty} \frac{1}{n+1} \cdot \frac{e^{-\lambda} \lambda^n}{n!}$$

$$\sum_{n=0}^{\infty} \frac{\lambda^n}{n!} = e^{\lambda}$$

$$= \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n+1)!} = \sum_{n=1}^{\infty} \frac{e^{-\lambda}}{\lambda} \cdot \frac{\lambda^n}{n!} = \frac{e^{-\lambda}}{\lambda} (e^{\lambda} - 1)$$

$$\Rightarrow P(X=n|Y=1) = \frac{\frac{1}{n+1} \frac{e^{-\lambda} \lambda^n}{n!}}{\frac{e^{-\lambda}}{\lambda} (e^{\lambda} - 1)} = \frac{\lambda^{n+1}}{(e^{\lambda} - 1)(n+1)!}$$

10. Gubner Chapter 2, 37

$X \sim \text{Ber}(p)$ and $Y \sim \text{Ber}(q)$ are independent

$$E[(X+Y)^3]$$

$$P(X=0)=1-p, P(X=1)=p, P(Y=0)=1-q, P(Y=1)=q.$$

$$E[(X+Y)^3] = \sum_{\substack{x \in \{0,1\} \\ y \in \{0,1\}}} (x+y)^3 P(X=x, Y=y) = \sum_{\substack{x \in \{0,1\} \\ y \in \{0,1\}}} (x+y)^3 P(X=x)P(Y=y)$$

$$= 0 \cdot P(X=0)P(Y=0) + 1^3 \cdot P(X=0)P(Y=1) + 1^3 \cdot P(X=1)P(Y=0) + 2^3 \cdot P(X=1)P(Y=1)$$

$$= 0 + (1-p)q + p \cdot (1-q) + 8pq$$

$$= 6pq + p + q$$

$$(X+Y)^3 = X^3 + 3X^2Y + 3XY^2 + Y^3 = X + 3XY + 3XY + Y = X + Y + 6XY$$

$$\therefore E[(X+Y)^3] = E(X) + E(Y) + 6E(XY)$$

11. Simulation problem 1a using R for $p=0.5$ and $\lambda=1$